# THE SERENDIB UNIVERSITY ENDOWMENT FUND — EXECUTIVE TEAR SHEET

USD 10 Million Investment Mandate Defense | Quantitative Risk & Extreme Value Theory Engine

**Live Cloud App:** quant-edge-final-challenge-vnzkxsfkkownmkpp62jboj.streamlit.app • **Code Repository:** github.com/Sahas-sass/quant-edge-final-challenge

**OFFICIAL RECOMMENDATION: ACCEPT MANDATE WITH EVT TAIL-RISK BUFFERS**

**CAPITAL PRESERVATION SCORE: EXCELLENT**

The research division recommends accepting the USD 10M endowment mandate under the proposed **Conservative Endowment Allocation**. The portfolio achieves **+7.89% annualized return** (exceeding the university's CPI+4% spending hurdle) with low volatility (**10.40%**), superior drawdown protection (**-28.5%** vs **-34.7%** for 60/40), and rigorous tail defense calibrated via Extreme Value Theory (EVT).

**Mandate Size:** $10,000,000 USD
**Horizon:** Perpetual / Long-Term
**Regulatory Status:** Basel GREEN Zone
**EVT 99% 1D ES:** 2.75% ($275,160)

## 1. Recommended Asset Allocation vs Standard Benchmarks

| Portfolio | Allocation Weights (%) | Ann. Return | Ann. Volatility | Sharpe (Rf=2%) | Sortino | Max Drawdown | CAPM Beta |
|---|---|---|---|---|---|---|---|
| **Conservative Endowment** | AGG:40%, SPY:20%, GLD:15%, EFA:10%, XLE:5%, XLK:5%, EEM:5% | **7.89%** | 10.40% | **0.61** | **0.87** | **-28.52%** | 0.48 |
| Standard 60/40 Benchmark | SPY: 60%, AGG: 40% | 8.55% | 11.93% | 0.59 | 0.83 | -34.70% | 0.60 |
| Equal Weight (1/N) | All 7 assets at 14.3% each | 10.24% | 16.28% | 0.53 | 0.76 | -42.35% | 0.77 |

## 2. Risk Dashboard: VaR vs Expected Shortfall vs Extreme Value Theory (EVT)

| Confidence Level | Parametric VaR | Historical VaR | EVT-POT VaR | Parametric ES | Historical ES | EVT-POT ES | Tail Risk Gap (%) |
|---|---|---|---|---|---|---|---|
| **90% Confidence** | 0.81% | 0.64% | 0.64% | 1.12% | 1.15% | **1.15%** | +3.2% |
| **95% Confidence** | 1.05% | 0.94% | 0.93% | 1.32% | 1.54% | **1.54%** | +17.0% |
| **99% Confidence** | 1.49% | 1.84% | **1.84%** | 1.71% | 2.76% | **2.75%** | **+60.5%** |

## 3. Crisis Stress Testing on USD 10 Million Portfolio

| Crisis Scenario | Market Episode / Narrative | Portfolio Return | Max Drawdown | Dollar Impact ($USD) | Capital Preserved |
|---|---|---|---|---|---|
| **2008 GFC Lehman Shock** | Sep 1, 2008 – Nov 20, 2008 (Global banking panic) | -21.40% | -22.64% | **-$2,139,553** | $7,860,447 |
| **March 2020 COVID Crash** | Feb 19, 2020 – Mar 23, 2020 (Global liquidity crunch) | -16.98% | -19.07% | **-$1,698,423** | $8,301,577 |
| **2022 Fed Rate-Hiking Drawdown** | Jan 3, 2022 – Oct 14, 2022 (500bps rate shock) | -15.94% | -16.55% | **-$1,594,042** | $8,405,958 |
| **Custom 2026-27 Oil/Rate Shock** | Oil +60%, Rates high, Tech -32%, Gold +22% | **-10.75%** | -10.75% | **-$1,075,000** | **$8,925,000** |

## 4. Key Takeaways for the Investment Committee

1. **Superior Risk-Adjusted Profile:** Generates +7.89% return (Sharpe 0.61, Sortino 0.87) vs +8.55% for 60/40, while cutting max drawdown from -34.7% down to -28.5%.
2. **Why EVT is Crucial:** Conventional normal models understate 99% tail loss by **+60.5%**. EVT accurately models fat tails (GPD shape $\xi$=0.244).
3. **Proven Backtest Reliability:** Passed out-of-sample testing with 16 hits vs 17 expected (Kupiec p=0.8149), placing the fund in the **Basel GREEN Zone**.